

SYDNEY TECHNICAL HIGH SCHOOL



HIGHER SCHOOL CERTIFICATE ASSESSMENT TASK 2 MARCH 2016

Mathematics Extension 2

Name

Teacher

General Instructions

- Reading Time - 5 minutes.
- Working Time - 90 min.
- Write using a blue or black pen.
- Board approved calculators may be used.
- The BOSTES reference sheet is provided.
- In Questions 7-10, show relevant mathematical reasoning and /or calculations.
- Begin each question on a fresh sheet of paper.
- Full marks may be not be awarded for careless and illegible writing.

Total marks (68)

- Attempt Questions 1-10.

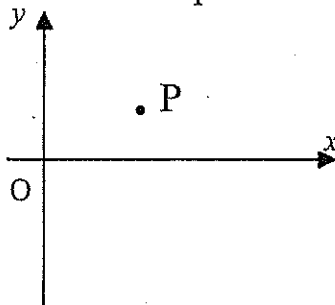
Multiple Choice		6
Question 7		16
Question 8		14
Question 9		14
Question 10		16
TOTAL		66

Section 1

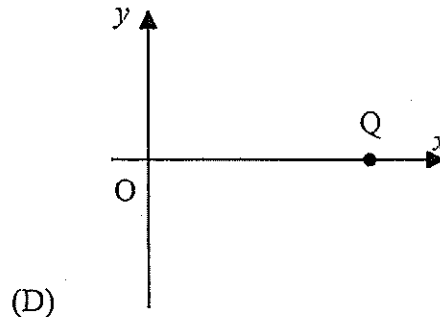
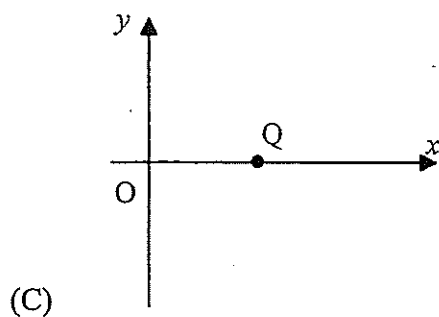
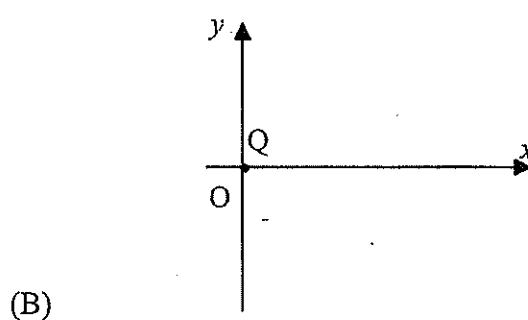
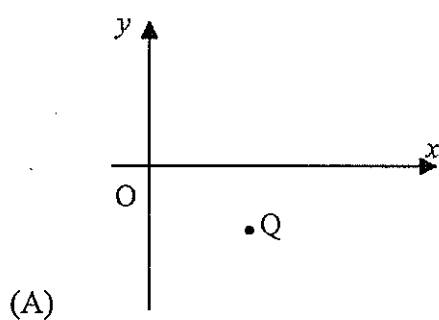
Multiple Choice (6 marks)

Use the multiple choice answer sheet for Question 1-5

1. In the Argand Diagram below, P represents the complex number z .



Which of the following Argand diagram shows the point Q representing $z + \bar{z}$?



2. What is the acute angle between the asymptotes of the hyperbola $\frac{x^2}{3} - y^2 = 1$

(A) $\frac{\pi}{6}$

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{3}$

(D) $\frac{\pi}{2}$

3. Which conic has eccentricity $\frac{\sqrt{13}}{3}$?

(A) $\frac{x^2}{3} + \frac{y^2}{2} = 1$

(B) $\frac{x^2}{3^3} + \frac{y^2}{2^2} = 1$

(C) $\frac{x^2}{3} - \frac{y^2}{2} = 1$

(D) $\frac{x^2}{3^3} - \frac{y^2}{2^2} = 1$

4. The equation $x^2 - xy + y^2 = 3$ defines y implicitly in terms of x .

The expression for $\frac{dy}{dx}$ is:

(A) $\frac{y - 2x + 3}{2y - x}$

(B) $\frac{y - 2x + 3}{x - 2y}$

(C) $\frac{y - 2x}{2y - x}$

(D) $\frac{y - 2x}{x - 2y}$

5. Consider the polynomial $P(x)$ of degree 3.

Two real numbers $a < b$ are such that:

$$a < b$$

$$P(a) > P(b) > 0$$

$$P'(a) = P'(b) = 0$$

The polynomial has:

- (A) 3 real zeros
- (B) 1 real zero γ such that $\gamma < a$
- (C) 1 real zero γ such that $a < \gamma < b$
- (D) 1 real zero γ such that $\gamma > b$

6. $P(4,25)$ is a point on the rectangular hyperbola $xy = 100$.

The tangent at P cuts the hyperbola's asymptotes at Q and R.

The area of $\triangle OQR$ (where O is the origin) is:

(A) $200\sqrt{2} u^2$

(B) $2\sqrt{50} u^2$

(C) $100 u^2$

(D) $200 u^2$

Section II

Total Marks (64)

Attempt Questions 7 – 10.

Answer each question in your writing booklet.

In Questions 6-9, your responses should include relevant mathematical reasoning and/or calculations.

Question 7 (16 Marks)

Use a Separate Sheet of paper

- a) Consider the complex numbers $z = -\sqrt{3} + i$ and $w = 3\left(\cos \frac{\pi}{7} + i \sin \frac{\pi}{7}\right)$
- i) Evaluate $|z|$ 1
 - ii) Evaluate $\arg(z)$ 1
 - iii) Find the argument of $\frac{z}{w}$ 1
- b) Find the value of $\frac{dy}{dx}$ at the point $(2, -1)$ on the curve $x + x^2 y^3 = -2$ 3
- c) Find the Cartesian equation of the locus of a point P which represents the complex number z where $|z - 2i| = |z|$. 2
- d) Let $w = 3 - 4i$ and $z = 2 + 2i$.
- i) Find $w\bar{z}$ in the form $x + iy$ 1
 - ii) Find $\operatorname{Im}\left(\frac{1}{2 - w}\right)$ 2
 - iii) The point representing the complex number z is rotated 270° in an anticlockwise direction about the origin in an Argand diagram.
What is the complex number represented by the new position of the point? 1
- e) i) Sketch and shade on an Argand diagram the region R in which $|z - 4i| \leq 3$ and $0 \leq \arg(z + 1) \leq \frac{\pi}{4}$ hold simultaneously. 2
- ii) Find the value of $\arg(z + 1)$ at the point in the region R where $\arg(z + 1)$ is a minimum. Answer to the nearest degree. 2

End of Question 7

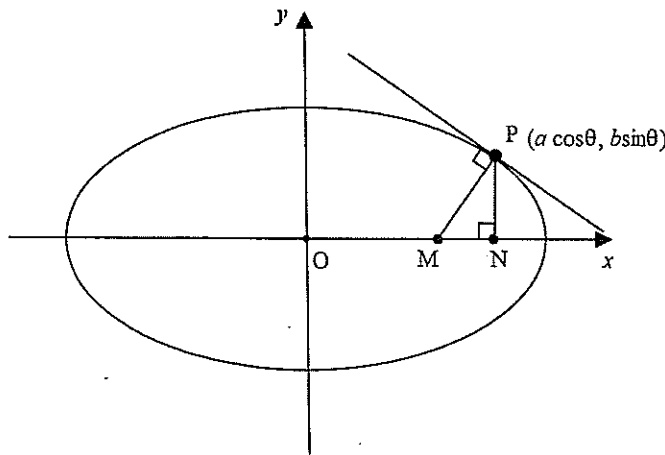
Question 8 (14 Marks)

Use a Separate Sheet of paper

a) Express $\frac{7+2i}{5-i}$ in the form $x+iy$ where x and y are real 2

- b) If $z = (1-i)^{-1}$
- i) Express $\bar{z}$ in modulus argument form 2
- ii) If $(\bar{z})^{13} = a+ib$ where a and b are real numbers, find the values of a and b 2

- c) $P(a \cos \theta, b \sin \theta)$ is a point on the ellipse with equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$
 The normal at P meets the x -axis at M and PN is perpendicular to the x -axis.



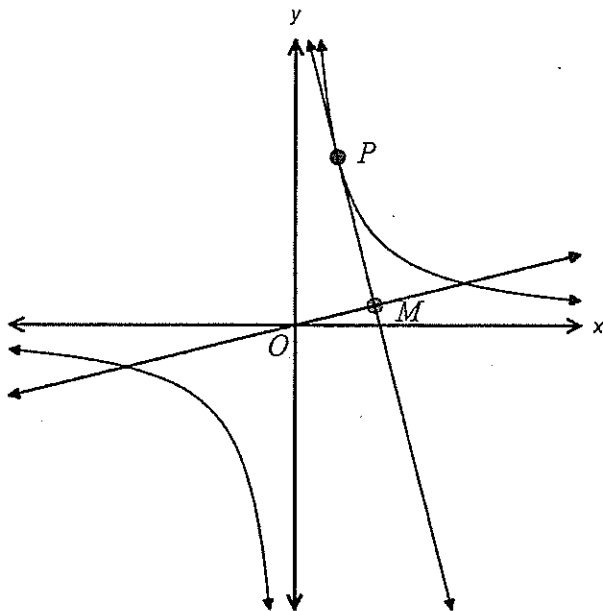
- i) Show that the equation of the normal at P is given by $ax \sin \theta - by \cos \theta = (a^2 - b^2) \sin \theta \cos \theta$ 2
- ii) Hence show that $MN = \left| \frac{b^2 \cos \theta}{a} \right|$. 2
- d) i) If a is a root of $P(x)$ with multiplicity n , show that a is also a root of $P'(x)$ with multiplicity $n-1$. 1
- ii) Given $P(x) = 2x^4 + 9x^3 + 6x^2 - 20x - 24$ has a triple root, factorise $P(x)$ into its linear factors. 3

End of Question 8

Question 9 (14 Marks)

Use a Separate Sheet of paper

- a) $P\left(t, \frac{1}{t}\right)$ is a variable point on the rectangular hyperbola $xy = 1$. M is the foot of the perpendicular from the origin to the tangent to the hyperbola at P.



- i) Show that the tangent to the hyperbola at P has equation $x + t^2y = 2t$ 2
- ii) Find the equation of OM 1
- iii) Show that the equation of the locus of M as P varies is $x^4 + 2x^2y^2 - 4xy + y^4 = 0$ and indicate any restrictions on the values of x and y . 3
- b) The polynomial $p(x) = x^5 + 2x^2 + mx + n$ has a double zero at $x = -2$. Find the value of m and n , and find the product of the other three zeros. 3
- c) Sketch the hyperbola with parametric equations
- $$x = 3 \sec \theta$$
- $$y = 4 \tan \theta$$

Indicate the vertices, the foci, and the equations of the directrices and asymptotes

5**End of Question 9**

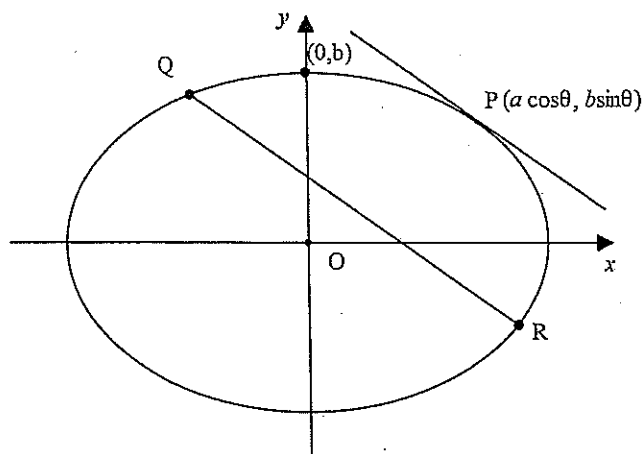
Question 10 (16 Marks)

Use a Separate Sheet of paper

- a) Sketch the region in the complex plane where $\operatorname{Re}[(2 - 3i)z] < 12$

2

- b) Consider the ellipse $\mathcal{E}$, with equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ and the points $P(a \cos \theta, b \sin \theta)$, $Q(a \cos(\theta + \varphi), b \sin(\theta + \varphi))$ and $R(a \cos(\theta - \varphi), b \sin(\theta - \varphi))$ on $\mathcal{E}$.



- i) Show that the equation of the tangent to $\mathcal{E}$ at the point P is $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$. 2
- ii) Show that the chord QR is parallel to the tangent at P. 2
- iii) Show that OP bisects the chord QR. 3
- c) i) Using De Moivre's theorem show that the solution of the equation $z^3 = 1$ in the complex number system are:
 $z = \cos \theta + i \sin \theta$ for $\theta = 0, \frac{2\pi}{3}, \frac{4\pi}{3}$ 2
- ii) If $\omega = \operatorname{cis} \frac{2\pi}{3}$ show that $\omega^2 + \omega + 1 = 0$ and $\omega^3 - \omega^2 - \omega - 2 = 0$ 2
- iii) Hence or otherwise solve the cubic equation $z^3 - z^2 - z - 2 = 0$ over the complex field 3

End of Examination

Extension 2 Mathematics
Assessment Task 2
March 2016

Multiple Choice

1. D 2. C 3. D 4. C 5. B 6. D

Question 7

a.i) $|z| = -\sqrt{3} + i$ $w = 3(\cos \frac{\pi}{7} + i \sin \frac{\pi}{7})$
 $|z| = |\sqrt{3} + i|$
 $= \sqrt{(\sqrt{3})^2 + 1}$
 $= 2$

ii) $\arg(z) = \alpha$ (z is in 2nd Quad)
 $\tan \alpha = -\frac{1}{\sqrt{3}}$
 $\arg(z) = \tan^{-1}(-\frac{1}{\sqrt{3}})$
 $= \pi - \frac{\pi}{6}$
 $= \frac{5\pi}{6}$

iii) $\arg(\frac{z}{w}) = \frac{5\pi}{6} - \frac{\pi}{7}$
 $= \frac{29\pi}{42}$

b) $x^2 + x^2 y^3 = -2$
 $1 + 2xy^3 + x^2 \cdot 3y^2 \frac{dy}{dx} = 0$
 $3x^2 y^2 \frac{dy}{dx} = -1 - 2xy^3$
 $\frac{dy}{dx} = \frac{-1 - 2xy^3}{3x^2 y^2}$

At (2, -1) $\frac{dy}{dx} = \frac{1}{4}$

b) $x + x^2 y^3 = -2$
 $1 + 2xy^3 + x^2 \cdot 3y^2 \frac{dy}{dx} = 0$
 $3x^2 y^2 \frac{dy}{dx} = -1 - 2xy^3$
 $\frac{dy}{dx} = \frac{-1 - 2xy^3}{3x^2 y^2}$

At (2, -1) $\frac{dy}{dx} = \frac{1}{4}$

c) $|z - 2i| = |z|$

perpendicular bisector of (0,0) and (0,2)

$\therefore$ The locus is the line $y=1$

di) $w = 3 - 4i$ $z = 2 + 2i$
 $w\bar{z} = (3 - 4i)(2 - 2i)$
 $= 6 - 6i - 8i + 8i^2$
 $= -2 - 14i$

ii) $2 - w = 2 - (3 - 4i)$
 $= -1 + 4i$

$\frac{1}{2-w} = \frac{1}{-1+4i} \times \frac{-1-4i}{-1-4i}$

$= \frac{-1-4i}{1-16i^2}$

$= \frac{-1-4i}{17}$ $\therefore \operatorname{Im}\left(\frac{1}{2-w}\right) = \frac{-4}{17}$

c)

y

$$|z - 2i| = |z|$$

perpendicular bisector of
(0,0) and (0,2)

x

∴ The locus is the line $y=1$

d) $w = 3 - 4i$ $z = 2 + 2i$

$$\begin{aligned} \text{i) } w\bar{z} &= (3 - 4i)(2 - 2i) \\ &= 6 - 6i - 8i + 8i^2 \\ &= -2 - 14i \end{aligned}$$

$$\begin{aligned} \text{ii) } 2 - w &= 2 - (3 - 4i) \\ &= -1 + 4i \end{aligned}$$

$$\frac{1}{2 - w} = \frac{1}{-1 + 4i} \times \frac{-1 - 4i}{-1 - 4i}$$

$$= \frac{-1 - 4i}{1 - 16i^2}$$

$$= \frac{-1 - 4i}{17}$$

$$\therefore \operatorname{Im}\left(\frac{1}{2 - w}\right) = -\frac{4}{17}$$

$$\text{ii) Multiplying } i^3 = 270^\circ$$

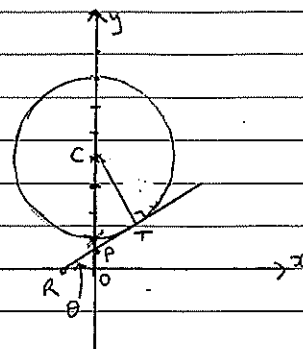
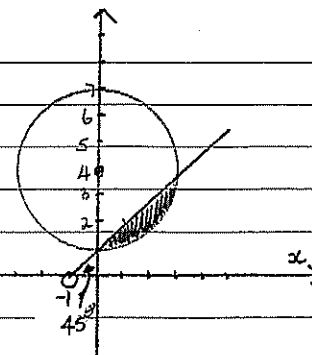
$$= i^3(z + 2i)$$

$$= -i(2 + 2i)$$

$$= -2i - 2i^2$$

$$= -2i + 2$$

2.



$$RO = 1$$

$$CT = 3$$

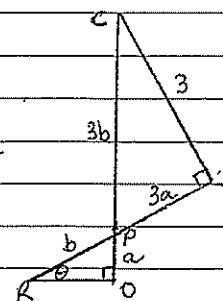
$$PO = a$$

$$PR = b$$

$$CO = 4 \text{ units}$$

$$\therefore \triangle CTP \parallel \triangle OPR$$

$$1 : 3$$



$$1^2 + a^2 = b^2$$

$$a + 3b = 4$$

$$a = 4 - 3b$$

$$1^2 - a^2 = b^2$$

$$1 + (4 - 3b)^2 = b^2$$

$$1 + 16 - 24b + 9b^2 = b^2$$

$$8b^2 - 24b + 17 = 0$$

$$b = \frac{24 \pm \sqrt{24^2 - 4 \times 8 \times 17}}{2 \times 8}$$

$$b = \frac{24 \pm \sqrt{32}}{16} \Rightarrow b = \frac{6 \pm \sqrt{2}}{4}$$

$$\cos \theta = 1 \div \left(\frac{6 \pm \sqrt{2}}{4}\right)$$

$$\cos \theta = 0.87, 0.539$$

$$\theta = 29^\circ, 57^\circ$$

Minimum value of $\arg(z+1)$
is 29°

Question 8

$$\begin{aligned} \text{a) } \frac{7+2i}{5-i} \times \frac{5+i}{5+i} &= \frac{35+7i+10i+2i^2}{25+1} \\ &= \frac{33+17i}{26} \\ &= \frac{33}{26} + \frac{17}{26}i \end{aligned}$$

$$\begin{aligned} \text{b) i) } z &= \frac{1}{1-i} \times \frac{1+i}{1+i} = \frac{1+i}{1-i^2} \\ \bar{z} &= \frac{1}{2} - \frac{1}{2}i \end{aligned}$$

$$|\bar{z}| = \sqrt{\left(\frac{1}{2}\right)^2 + \left(-\frac{1}{2}\right)^2} = \frac{1}{\sqrt{2}}$$

$$\arg(\bar{z}) = -\frac{\pi}{4}$$

$$\therefore \bar{z} = \frac{1}{\sqrt{2}} \operatorname{cis}\left(-\frac{\pi}{4}\right)$$

$$\text{ii) } (\bar{z})^{13} = \left[\frac{1}{\sqrt{2}} \operatorname{cis}\left(-\frac{\pi}{4}\right) \right]^{13}$$

$$= \left(\frac{1}{\sqrt{2}}\right)^{13} \operatorname{cis}\left(-\frac{13\pi}{4}\right)$$

$$= \frac{1}{64\sqrt{2}} \operatorname{cis}\left(\frac{3\pi}{4}\right)$$

$$= \frac{1}{64\sqrt{2}} \left(-\frac{1}{2} + \frac{1}{2}i\right)$$

$$a = -\frac{1}{128} \quad b = \frac{1}{128}$$

$$\text{c) i) } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} + \frac{2y}{b^2} \times \frac{dy}{dx} = 0$$

$$\begin{aligned} \frac{dy}{dx} &= -\frac{2x}{a^2} \times \frac{b^2}{2y} \\ &= -\frac{xb^2}{a^2y} \end{aligned}$$

$$\text{At } P(a \cos \theta, b \sin \theta) \quad \frac{dy}{dx} = -\frac{b^2 a \cos \theta}{a^2 (b \sin \theta)}$$

$$\text{Gradient of tangent at } P = -\frac{b \cos \theta}{a \sin \theta}$$

$$\text{Gradient of normal at } P = \frac{a \sin \theta}{b \cos \theta}$$

Equation of normal at P

$$y - b \sin \theta = \frac{a \sin \theta}{b \cos \theta} (x - a \cos \theta)$$

$$yb \cos \theta - b^2 \sin \theta \cos \theta = a \sin \theta x - a^2 \sin \theta \cos \theta$$

$$ax \sin \theta - by \cos \theta = (a^2 - b^2) \sin \theta \cos \theta$$

$$\text{cii) } N(a \cos \theta, 0)$$

let $y=0$ equation of normal

$$ax \sin \theta = (a^2 - b^2) \sin \theta \cos \theta$$

$$x = \frac{(a^2 - b^2) \sin \theta \cos \theta}{a \sin \theta}$$

$$\therefore M = \left[\frac{(a^2 - b^2) \cos \theta}{a}, 0 \right]$$

$$MN = \left| \frac{(a^2 - b^2) \cos \theta - a \cos \theta}{a} \right|$$

$$= \left| \frac{a^2 \cos \theta - b^2 \cos \theta - a^2 \cos \theta}{a} \right|$$

$$= \left| \frac{b^2 \cos \theta}{a} \right|$$

d) i) $P(x) = (x - \alpha)^n Q(x) \quad Q(\alpha) \neq 0$

$$P'(x) = n(x - \alpha)^{n-1} Q(x) + (x - \alpha)^n \cdot Q'(x)$$

$$= (x - \alpha)^{n-1} [n Q(x) + (x - \alpha) \cdot Q'(x)]$$

$$= (x - \alpha)^{n-1} \cdot Q(x) \quad \text{where } Q(\alpha) \neq 0$$

$\therefore \alpha$ is a root of $P'(x)$ of multiplicity $n-1$

die) $P(x) = 2x^4 + 9x^3 + 6x^2 - 20x - 24$

$$P'(x) = 8x^3 + 27x^2 + 12x - 20$$

$$P''(x) = 24x^2 + 54x + 12$$

$$\therefore 24x^2 + 54x + 12 = 0$$

$$4x^2 + 9x + 2 = 0$$

$$(4x+1)(x+2) = 0$$

$$x = -\frac{1}{4} \quad x = -2$$

Sub $x = -2 \quad P(-2) = 2(-2)^4 + 9(-2)^3 + 6(-2)^2 - 20(-2) - 24 = 0$

$\therefore x = -2$ is a triple root

$$\therefore P(x) = (x-2)^3(2x-3)$$

Question 9

ai) $\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$

$$= -\frac{1}{t^2} \times 1$$

$$= -\frac{1}{t^2}$$

$$y - \frac{1}{t} = -\frac{1}{t^2} (x - t)$$

$$t^2 y - t = -x + t$$

$$\therefore x + t^2 y = 2t$$

ii) $y = t^2 x$ Gradient of OM is perpendicular to tangent at P.

iii) $t^2 = \frac{y}{x}$

$$t = \pm \sqrt{\frac{y}{x}}$$

$$x + t^2 y = 2t$$

$$x + \frac{y}{x} \times y = 2 \times \pm \sqrt{\frac{y}{x}}$$

$$x + \frac{y^2}{x} = 2 \times \left(\pm \sqrt{\frac{y}{x}} \right)$$

$$\left(x + \frac{y^2}{x} \right)^2 = 4 \times \left(\frac{y}{x} \right)$$

$$x^2 + 2y^2 + \frac{y^4}{x^2} = \frac{4y}{x}$$

$$x^4 + 2x^2 y^2 + y^4 = 4xy$$

$\rightarrow$ restriction $\frac{y}{x} > 0$

The point M cannot lie on the hyperbola

$$M\left(\frac{2t}{1+t^4}, \frac{2t^3}{1+t^4}\right)$$

(Alternate Method)

$$x = \frac{2t}{1+t^4} \quad y = \frac{2t^3}{1+t^4}$$

$$x^4 + 2x^2y^2 - 4xy + y^4 = 0$$

$$\left(\frac{2t}{1+t^4}\right)^4 + 2\left(\frac{2t}{1+t^4}\right)^2\left(\frac{2t^3}{1+t^4}\right)^2 - 4\left(\frac{2t}{1+t^4}\right)\left(\frac{2t^3}{1+t^4}\right) + \left(\frac{2t^3}{1+t^4}\right)^4 = 0$$

$$\frac{16t^4}{(1+t^4)^4} + \frac{32t^8}{(1+t^4)^4} - \frac{16t^4}{(1+t^4)^2} + \frac{16t^{12}}{(1+t^4)^4}$$

$$\frac{16}{(1+t^4)^4} [t^4 + 2t^8 + t^{12}] - \frac{16t^4}{(1+t^4)^2}$$

$$\frac{16}{(1+t^4)^4} (t^4 + 2t^8 + t^{12}) - \frac{16t^4(1+2t^4+t^8)}{(1+t^4)^4}$$

$$\frac{16}{(1+t^4)^4} [t^4 + 2t^8 + t^{12} - (t^4 + 2t^8 + t^{12})]$$

$$\frac{16}{(1+t^4)^4} \times 0$$

$$= 0$$

$$\therefore \text{locus of } P \text{ is } x^4 + 2x^2y^2 - 4xy + y^4 = 0$$

$$P(t, \frac{1}{t}) \therefore t \neq 0$$

M cannot be at the origin

b)

$$P(x) = x^5 + 2x^2 + mx + n$$

$$P'(x) = 5x^4 + 4x + m$$

$P(x)$ has a double root at $x = -2$

$$P'(-2) = 0$$

$$80 - 8 + m = 0$$

$$m = -72$$

$$P(-2) = 0$$

$$-32 + 8 + 144 + n = 0$$

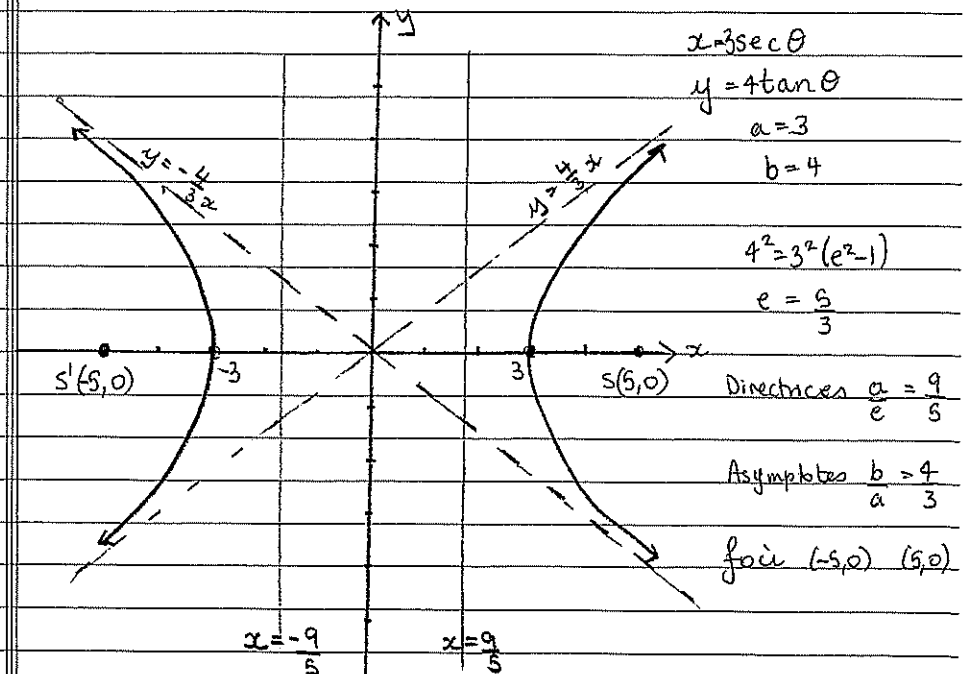
$$n = -120$$

Roots : $-2, -2, \alpha, \beta, \gamma$

$$\text{Product of roots } 4\alpha\beta\gamma = 120$$

$$\alpha\beta\gamma = 30$$

d)



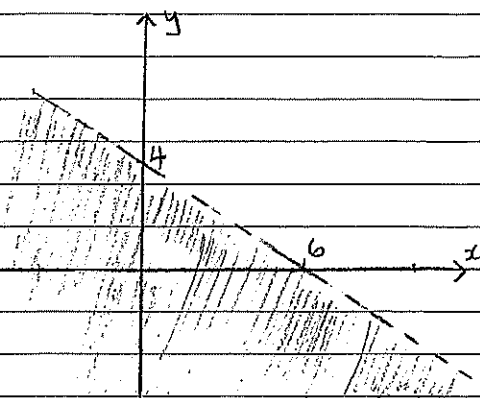
Question 10

a) $[(2-3i)z] < 12$

$$\operatorname{Re}[(2-3i)(x+iy)] < 12$$

$$\operatorname{Re}[2x+3y+i(2y-3x)] < 12$$

ie $2x+3y < 12$



b) $x = a \cos \theta$ $\frac{dx}{d\theta} = -a \sin \theta$
 $y = b \sin \theta$ $\frac{dy}{d\theta} = b \cos \theta$

$$\frac{dy}{dx} = \frac{dy}{d\theta} \div \frac{dx}{d\theta}$$

$$= b \cos \theta \times \frac{1}{-a \sin \theta}$$

$$= \frac{-b \cos \theta}{a \sin \theta}$$

Equation of tangent at P

$$y - b \sin \theta = \frac{-b \cos \theta}{a \sin \theta} (x - a \cos \theta)$$

$$y a \sin \theta - a b \sin^2 \theta = -x b \cos \theta + a b \cos^2 \theta$$

$$x b \cos \theta + y a \sin \theta = a b (\cos^2 \theta + \sin^2 \theta)$$

$$x b \cos \theta + y a \sin \theta = a b$$

$$\frac{x b \cos \theta}{a b} + \frac{y a \sin \theta}{a b} = 1$$

$$\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1$$

bii) Gradient of QR

$$Q [a \cos(\theta+\phi), b \sin(\theta+\phi)] \quad R [a \cos(\theta-\phi), b \sin(\theta-\phi)]$$

$$M_{QR} = \frac{b \sin(\theta-\phi) - b \sin(\theta+\phi)}{a \cos(\theta-\phi) - a \cos(\theta+\phi)}$$

$$= \frac{b(\sin \theta \cos \phi - \cos \theta \sin \phi - \sin \theta \cos \phi - \cos \theta \sin \phi)}{a(\cos \theta \cos \phi + \sin \theta \sin \phi - \cos \theta \cos \phi + \sin \theta \sin \phi)}$$

$$= \frac{-2b \cos \theta \sin \phi}{2a \sin \theta \sin \phi}$$

$$M_{QR} = \frac{-b \cos \theta}{a \sin \theta}$$

Gradient of tangent at P = $\frac{-b \cos \theta}{a \sin \theta}$

∴ chord QR is parallel to the tangent at P

biii) Let M be the midpoint of QR

$$\left[\frac{a \cos(\theta+\phi) + a \cos(\theta-\phi)}{2}, \frac{b \sin(\theta+\phi) + b \sin(\theta-\phi)}{2} \right]$$

$$x = \frac{a(\cos \theta \cos \phi - \sin \theta \sin \phi + \cos \theta \cos \phi + \sin \theta \sin \phi)}{2} = a \cos \theta \cos \phi$$

$$y = \frac{b(\sin \theta \cos \phi + \cos \theta \sin \phi + \sin \theta \cos \phi - \cos \theta \sin \phi)}{2} = b \sin \theta \cos \phi$$

$$M(a \cos \theta \cos \phi, b \sin \theta \cos \phi)$$

Equation of OP is $y = \frac{b \sin \theta}{a \cos \theta} x$

$$y = \frac{b \sin \theta}{a \cos \theta} \times a \cos \theta \cos \phi$$

$$y = b \sin \theta \cos \phi$$

M lies on OP

$\therefore$ OP bisects QR

civ) $z = \cos \theta + i \sin \theta$
 $- r^3 (\cos \theta + i \sin \theta)^3$
 $= r^3 (\cos 3\theta + i \sin 3\theta)$

$$r^3 = 1$$

$$\cos 3\theta + i \sin \theta = \cos 0 + i \sin 0$$

$$\cos 3\theta = \cos 0 \quad \sin 3\theta = \sin 0$$

$$\cos 3\theta = 1$$

$$3\theta = 0 + 2k\pi$$

$$\theta = \frac{0 + 2k\pi}{3} \quad k = 0, 1, 2$$

$$\theta = 0, \frac{2\pi}{3}, \frac{4\pi}{3}$$

civ) $\omega = \cos \frac{2\pi}{3}$ If ω is a root then $\omega^3 = 1$

$$\omega^3 - 1 = 0$$

$$(\omega - 1)(\omega^2 + \omega + 1) = 0$$

$$\omega^2 + \omega + 1 = 0$$

$$-\omega^2 - \omega - 1 = 0$$

$$\omega^3 - \omega^2 - \omega - 1 = 1$$

$$\omega^3 - \omega^2 - \omega - 2 = 0$$

civ)

$$z^3 - z^2 - z - 2 = 0$$

$$P(z) = (z)^3 - (z)^2 - (z) - 2$$

$$P(2) = 0$$

$\therefore z = 2$ is a root

$$\begin{array}{r} z^2 + z + 1 \\ z - 2 \overline{) z^3 - z^2 - z - 2} \\ \underline{z^3 - 2z^2} \\ z^2 - z \\ \underline{z^2 - 2z} \\ z - 2 \\ \underline{z - 2} \\ 0 \end{array}$$

$$P(z) = (z - 2)(z^2 + z + 1)$$

$$z^2 + z + 1 = 0$$

$$z = \frac{-1 \pm \sqrt{1 - 4}}{2}$$

$$z = \frac{-1 \pm \sqrt{3}i}{2}$$

$$z = \frac{-1 + \sqrt{3}i}{2}$$

$$z = \frac{-1 - \sqrt{3}i}{2}$$